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Instructions for Operation of the Model SD3 SpectraDrive Controller

Serial Number:

Ver. 1.0

SD3 SpectraDrive Controller

Description:

The SD3 SpectraDrive Controller is used with the Acton series AM and VM Monochromators in conjunction with Princeton Instruments Monochromator Control Software to provide the electronic drive for wavelength positioning, diverter mirror control and filter positioning. The SD3 also provides the link between the monochromator and the computer. Both USB 1.1 and RS-232 are supplied as standard interfaces.

Setup:

Each AM and VM monochromator ships with an SD3 controller and the software installation disk for Monochromator Control Software. Refer to the instruction manual for your specific AM or VM monochromator for unpacking and setup instructions. This manual is located on the Monochromator Control Software installation disk. After the monochromator is in position, grating installed and all packing materials removed, connect the monochromator control cable between the monochromator and the MONO connector on the rear panel of the SD3. Connect the power supply shipped with the SD3 to the POWER connector on the rear panel of the SD3. This power supply provides +5 volts DC for the SD3 control logic and +24 volts DC for the motor drives. With the power supply switch in the off position, connect the line cord provided with the SD3 to the power supply and the AC line. The power supply will operate with line voltages between 100 and 250 volts AC, 50/60 Hz. Use the Monochromator Control Software install disk to install the software on your computer. This icon for this software on your computer is labeled SpectraPro. Turn on the power switch on the SD3 power supply and the monochromator should initialize to zero wavelength.

Controlling the AM or VM Monochromator with Monochromator Control Software:

The Monochromator Control Software is normally installed in the directory C:\Program Files\SpectraPro. This directory contains subdirectories Bin and Data. The Bin directory contains the executable code. There will normally be a SpectraPro icon on the desktop which can be used for starting the software. If this icon is not on the desktop go to the SpectraPro Bin directory and start the software. When the Monochromator Control software loads, there is a main screen with selections for both operating the monochromator and various setup functions. Click on the Operation box and a screen will come up which allows for basic control of the monochromator wavelength. All functions of this software are described in the Monochromator Control Software manual supplied on the Monochromator Control Software install CD.

Control of the AM or VM Monochromator with User Generated Software or LabVIEW:

The Monochromator Control Software install disk contains a folder named SpectraPro_SDK. This folder contains dll's with example code in Delphi, Visual Basic, C++ and LabVIEW and well as a folder with examples for use of the COM interface.

Control of the AM or VM Monochromator at the Command Level:

Although it requires more programming on the user's part, the SD3 can also be controlled with direct commands through its USB 1.1 port or RS-232 port. The same command set, listed below, is used for both RS232 and USB.

Commands can be sent as single commands or grouped in strings of commands. All commands are single words (contain no spaces) and all commands in a string are separated by at least one space. Parameters, if needed, precede the command and are separated from the command by at least one space (e.g. 546.7 GOTO).

For RS232 operation, the port set-up is 9600 baud, 8 data bits, 1 stop bit and no parity. A convenient tool for trying out this mode of operation is the program HyperTerminal supplied with the Windows operating system. The USB 1.1 port with the driver supplied also shows up as and is treated like a com port – although a very fast one. All commands or strings of commands must be terminated with a carriage return (0D hex). The SD3 responds to a command when the command has been completed by returning the characters **OK** followed by carriage return and line feed (hex ASCII sequence 20 6F 6B 0D 0A). The default condition is to echo each character that is sent to the SD3 with the RS-232 interface and to not echo the commands when using the USB interface. When sending a command or string of commands, it is important to wait for the SD3 to complete the processing of that command string before sending another command.

Monochromator Wavelength Movement Commands:

GOTO	Goes to a destination wavelength at maximum motor speed. Accepts destination wavelength in nm as a floating point number with up to 4 digits after the decimal point or whole number wavelength with no decimal point.
<GOTO>	Same as GOTO (For compatibility with software written for previous Acton series models.)
NM	Goes to a destination wavelength at constant nm/min rate specified by last NM/MIN command. Accepts destination wavelength in nm as a floating point number with up to 4 digits after the decimal point or whole number wavelength with no decimal point.
<NM>	Same as NM (For compatibility with software written for previous Acton series models.)
>NM	Similar to NM except it returns control to user immediately rather than waiting for Completion of monochromator wavelength move. Can be used with ?NM or MONO-?DONE below. This command must be terminated with MONO-STOP listed below. NOTE: Use the NM command when communication with the monochromator during the scan is not required.
?NM	Returns present wavelength in nm to 0.01nm resolution with units nm appended. e.g. ?NM 300.00 nm
MONO-?DONE	Used with >NM command to determine if monochromator has reached the destination. Returns 0 if move is not complete, 1 if move is complete.
MONO-STOP	Stops the monochromator wavelength move after use of the >NM command.

NM/MIN	Sets constant scan rate in nm/min to 0.01 nm/min resolution. e.g. 10.0 NM/MIN
?NM/MIN	Returns present scan rate in nm/min to 0.01 nm/min resolution with units nm/min appended. e.g. ?NM/MIN 100.00 nm/min

Grating Control Commands:

GRATING	Recalls parameters for the specified grating from non-volatile memory. Up to nine (9) gratings are allowed. This command takes a grating number from 1 - 9. e.g. 3 GRATING
?GRATING	Returns the number of the grating presently being used numbered 1 - 9.
?GRATINGS	Returns the list of installed gratings with position, groove density and blaze. The present grating is specified with an arrow.

The following command is used for grating installation by Acton part #:

INSTALL	Installs new grating parameters into the non-volatile memory of the AM monochromator. Uses the part # of the grating to specify the parameters. e.g. 1-120-500 5 INSTALL places a 1200 g/mm grating blazed at 500nm into the second grating position on #5.
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The following commands are used for grating installation by grating parameters:

SELECT-GRATING	Specifies the grating number to be installed 1 - 9.
G/MM	Specifies groove density of grating to be installed in g/mm. e.g. 1200 G/MM
BLAZE	Specifies the blaze wavelength and units of the grating to be installed with 7 characters of the user's choice. Unlike other commands, this command is issued before the parameters. After the command is issued, the SD3 responds with " ". Seven characters are then entered (these may be numbers, letters, spaces or special characters).
UNINSTALL	Used to remove a grating and its parameters from the SD3 non-volatile memory e.g. 3 UNINSTALL

Diverter Control Commands:

EXIT-MIRROR	Designates the exit diverter mirror to receive the diverter control commands. This command is for AM monochromators which can accept two diverter mirrors. The AM monochromators will accept this command but it is not required in these monochromators.
ENT-MIRROR	Designates the entrance diverter mirror to receive the diverter control commands. This command is for AM monochromators which can accept two diverter mirrors.
FRONT	Moves the designated diverter mirror to position the beam to the front port position.
SIDE	Moves the designated diverter mirror to position the beam to the side port position.
?MIRROR	Returns the position of the designated diverter mirror with the responses "front" and "side".

?MIR Returns the position of the designated diverter mirror with the responses 0 for front and 1 for side.

CALIBRATION COMMANDS:

INIT-OFFSET Sets the offset value for the designated grating. Default values are 25600 for all gratings. The grating designator used with this command is grating# - 1.
e.g. 25590. 0 INIT-OFFSET for setting offset on grating #1.
NOTE: This command requires a decimal point after the offset value. For the new parameters of this command to take effect, the monochromator must be initialized with the MONO-RESET command or by turning the power off and back on.

INIT-GADJUST Sets grating adjustment value for the designated grating. Default values are 10000 for all gratings. The limits on the parameter for this command are +/- 1000 for all gratings. The grating designator used with this command is the grating # - 1.
e.g. 9993 1 INIT-GADJUST for setting gadjust on the second grating.
NOTE: This command is to maintain compatibility with previous Acton series applications. For new applications, use the INIT-SP300-GADJUST command below. No decimal point is used with this command. For the new parameters of this command to take effect, the monochromator must be initialized with the MONO-RESET command or by turning the power off and back on.

MONO-EESTATUS Returns setup and grating calibration parameters for all gratings.

RESTORE FACTORY SETTINGS Returns all parameters including grating calibration parameters to the original factory calibrated settings. NOTE: This command will overwrite any calibration parameters set by the user.

MONO-RESET Initializes AM monochromator. Necessary after using INIT-OFFSET, INIT-GADJUST or INIT-SP300-GADJUST.

HELLO Same as MONO-RESET. Used to maintain compatibility with existing applications.

MODEL Returns model number of AM or VM monochromator.
e.g. MODEL AM-505

SERIAL Returns serial number of AM or VM monochromator.
e.g. SERIAL 37480263

The following are the Start-Up parameters and their default values:

Default Values:

GRATING #1
WAVELENGTH 0.0 nm
SCAN SPEED 200.0 nm/min

INIT-GRATING Selects which of the three gratings on the installed in the AM monochromator will go to after finding 0.0 nm on the first grating of the installed turret. e.g. 2 INIT-GRATING selects the second grating as the default. Accepts values 1 - 9.

INIT-WAVELENGTH Sets an initial wavelength for the SD3 after initialization.
E.G. 435.84 INIT-WAVELENGTH

INIT-SRATE

Sets an initial scan rate for the SD3.
E.G. 200.0 INIT-SRATE

Cables and Connections for SpectraDrive-SD3

The following computer cables are supplied with the SD3:

CC-499-2 IBM PC or compatible 9 pin female (DB9S) connector to 9 pin male connector (DB9P).

3650-USB-06 USB Cable Type A to Type B.

If neither of these cables are compatible with your system, consult Princeton Instruments for a custom cable. If you have facilities for constructing a custom cable, use the RS232 pin arrangement shown in Table 1. See also Appendix B.

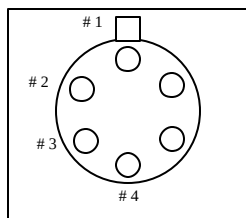
Table 1: RS232 Computer Interface Pin Arrangement:

pin #	description
1	open
2	RD data from SD3 to computer
3	TD data from computer to SD3
4	open
5	ground
6	open
7	RTS
8	CTS
9	open

A terminal or RS232 computer port must be set up as follows: 9600 baud, 8 data bits, no parity, 1 start bit, 1 stop bit.

Table 2: Power Input Pin Arrangement:

PIN#	DESCRIPTION
1	+5V
2	GND
3	GND
4	+24V



SD3 Cable Connections:

Figure below shows the cable connections necessary for operation from a computer through the COM ports or with USB. Refer to the Monochromator Control Software for Windows for operation with a computer.

